ENGINEERING DESIGN

OVERVIEW

Applying leadership and 21st century skills, participants use the engineering design process to research, design, build, test, and communicate a solution to the annual engineering design challenge theme posted on the TSA website under *Themes & Problems*. Through their documentation portfolio, display, prototype/model, and presentation, teams demonstrate their understanding of the design process and the application of their solution.

ELIGIBILITY

Five (5) teams of two to four (2-4) individuals per state may participate.

TIME LIMITS

A. Ten (10) minutes are allowed for the demonstration/presentation/interview.
   1. six (6) minutes are allowed for the team to demonstrate the solution and complete a presentation
   2. four (4) minutes are allowed for the team to respond to questions

ATTIRE

TSA competition attire is required for this event.

PROCEDURE

PRE-CONFERENCE

A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules.
B. Participants review the Use of Generative Artificial Intelligence (GenAI) found in the General Rules.
C. Participants access the annual theme posted on the TSA website under *Themes & Problems*.
D. Participants concentrate their efforts conducting research on engineering practices and brainstorming a solution.
E. Participants create and test a prototype/model of their solution.
F. Participants prepare their documentation and display according to the regulations.

PRELIMINARY ROUND

A. No more than two (2) team members report to the event area at the time and place stated in the conference program to check in:
   1. documentation portfolio
   2. a free-standing display
   3. a prototype/model of the solution
B. Entries are evaluated by the judges with neither students nor advisors present based on the following criteria:
   1. Judges score the Display and Prototype/Model criteria to determine the top twenty-four (24) preliminary contestants, which will not be posted.
   2. Judges score the Documentation Portfolio criteria of those top twenty-four (24) contestants to determine the top twelve (12) semifinalists.
C. A list of twelve (12) semifinalists (in random order) is posted.

SEMIFINAL ROUND

A. Participants report at the time and place stated in the conference program to sign up for a demonstration/presentation/interview time.
B. Participants report at the assigned time and place for the demonstration/presentation/interview.
C. A maximum of three (3) semifinalist team members demonstrate the accuracy of their model in solving the challenge and then present in front of their display and model/prototype, which may be used as a reference.
D. Judges evaluate the entries.
E. The top ten (10) finalists are announced at the awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRELIMINARY ROUND

A. Students prepare a documentation portfolio that includes each step of the engineering design process when developing a solution to the engineering challenge based on the annual theme.

B. Documentation Portfolio:

1. Documentation materials (comprising "a portfolio") are required and must be secured in a clear front report cover with the following single-sided, 8½" x 11" pages, in this order:
    a. Title page with the challenge listed, event title, the team identification number, the conference city and state, and the year; one (1) page
    b. Table of contents; pages as needed
    c. Identification and definition of problem; one (1) page
    d. Information gathering that explains the importance of developing a solution to the engineering challenge and how a solution would impact the lives of people. A concise historical perspective of the challenge must also be included; one (1) page
    e. Identification and explanation of three (3) possible solutions to the challenge must be included. For each possible solution presented, a concise narrative must be included that supports the plausibility of each solution based on a specific scientific, technical, and/or engineering concept; one (1) page per solution, three (3) pages total
    f. Of the three (3) possible solutions, select the most plausible solution and create a prototype/model. Provide an appropriate, specific, and descriptive, visual representation of the solution (ex. engineering drawings, schematic, flowchart, etc.); pages as needed.
    g. A written summary of the of the iteration process in the design of the prototype and the results of each test; at a minimum, four (4) pages describing the below points are required:
        i. Refinements of the prototype based on evaluation/testing conducted. If a solution cannot be evaluated/tested, write in narrative form a reflection of possible refinements that could be made to the chosen solution based on the testing means developed.
        ii. A reflection of the effectiveness of the selected solution and the testing means developed (i.e. did or would the tests developed actually prove that the solution is plausible?).
        iii. Describe any other issues found during the iteration process.
    h. Communication of the solution – a written summary detailing how the solution meets the annual theme; one (1) page
    i. References and resources page in a professional citation style of the competitors choosing. Some examples of professional citation styles include MLA, APA, Chicago, and IEEE; pages as needed
    j. Resources and AI Reflection (see 2027 & 2028 Forms Appendix); maximum of four (4) pages

C. Display:

1. A free-standing display must be used and the dimensions of the free-standing display may not exceed 15" deep x 2.5' wide x 4' high.
2. A tangible prototype/model must be included with the display and must physically fit within the display board dimensions.
3. If the display and/or prototype/model requires power, they must be powered by dry-cell batteries or photo-voltaic cells.
    a. Power supply must physically fit within the display board dimensions.
    b. All power must be switched off once the team has completed set-up.
    c. If teams want judges to activate any electronic device in their model/display, complete instructions must be provided to judges on how to power up the model/display.

4. No harmful or illegal substances are permitted. No viruses, live plants, or animals are permitted. No dangerous processes, experiments, and/or physical models may be displayed/demonstrated.

SEMIFINAL ROUND

A. Participants report at the time and place stated in the conference program to sign up for a demonstration/presentation/interview time.
B. Participants report at the assigned time and place for the demonstration/presentation/interview.
C. A maximum of three (3) semifinalist team members present in front of thier display and model/prototype, which may be used as a reference.
D. Ten (10) minutes are allowed for the demonstration/presentation/interview. Six (6) minutes are allowed for the demonstration/presentation and four (4) minutes to respond to questions.
E. Teams will be provided with an additional revealed performance parameter at the conference (e.g., a target distance, load, or constraint). Teams may use data documented in their portfolio to set up their demonstration.
F. Judges evaluate the entries.
G. The top ten (10) finalists are announced at the awards ceremony.

EVALUATION

PRELIMINARY ROUND

Tier 1

A. Display
B. Prototype/model

Tier 2

C. Documentation portfolio

SEMIFINAL ROUND

A. Demonstration/presentation/interview

Refer to the official rating form for more information.

GENERATIVE ARTIFICIAL INTELLIGENCE

Generative Artificial Intelligence (GenAI) is permitted for this event. Refer to the Use of Generative Artificial Intelligence (GenAI) for Competitive Events found in the General Rules and Regulations.

STEM INTEGRATION

This event aligns with the STEM (Science, Technology, Engineering, and Mathematics) educational standards.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Engineer
- Environmental scientist
- Health and safety specialist
- Manufacturing consultant
- Mechanical engineer

Participant/Team ID# _____

ENGINEERING DESIGN

2027 & 2028 OFFICIAL RATING FORM

HIGH SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

☐ TIER 1 – Display and Prototype/model is present
☐ TIER 2 – Documentation portfolio
☐ ENTRY NOT EVALUATED

TIER 1 – DISPLAY AND PROTOTYPE/MODEL (40 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Aesthetics (X1)	Display design is unattractive in appearance and shows a lack of understanding of graphic design principles.	Display design is somewhat attractive and shows an adequate understanding of the use of graphic design principles.	Display is of professional quality with an exemplary use of graphic design principles.	
Prototype/Model (X1)	Models are confusing and do not represent and/or support the proposed problem solution.	Models provide adequate representation and support of the proposed problem solution.	Models provide excellent representation and support of the proposed problem solution.	
Overall Impact (X2)	Display information and models do not detail or enhance the essential components of the team's problem identification and solution.	Display information and models somewhat detail and enhance the essential components of the team's problem identification and solution.	Display information and models greatly detail and enhance the essential components of the team's problem identification and solution.	
			TIER 1 – DISPLAY AND PROTOTYPE/MODEL SUBTOTAL (40 points)	

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

TIER 1 – SUBTOTAL (40 points)	

TIER 2 – DOCUMENTATION PORTFOLIO (110 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Portfolio Components (X1)	Portfolio is unorganized and/or is missing three (3) or more components.	Portfolio has most components and is generally organized per the order of the event guide listing; it has sufficient content.	All components are included in the portfolio; content and organization per the order of the event guide listing are excellent.	
Identification and Problem Definition (X1)	Problem is not clearly defined or communicated.	Problem is somewhat defined and communicated.	Problem is clearly written, concise, and well defined.	
Information Gathering (X1)	There is little evidence of research; there is a lack of understanding of the design challenge.	There is some evidence of research; an adequate understanding of the design challenge.	Thorough research is clearly evident with a firm understanding of the design challenge.	
Possible Solutions (X1)	A very brief explanation of the final solution is presented; there is a lack of creativity; descriptions are weak.	An adequate description of the solution is presented and supported by some amount of research and evidence; solution is somewhat creative.	Solution is supported by the research gathered and scientific and engineering evidence; solution is plausible and creative.	
Selected Solution (X2)	Solution conveys a sloppy design, and/or does not incorporate key elements in the engineering challenge; visual representations of the solutions are not appropriate or accurate and do not follow established conventions.	Solution incorporates most elements laid out in the engineering challenge; visual representations of the solutions are somewhat appropriate, accurate and loosely follow established conventions.	Solution exudes creativity and addresses all engineering challenge elements; visual representations of the solutions are appropriate, accurate and follow established conventions.	
Written Summary of Iteration Process (X2)	Summary has little support or evidence of the testing process of each stage of iteration.	Summary includes refinement and reflection provides evidence of multiple evaluations and testing.	Summary includes exemplary examples of refinement and reflection of the design process and details issues found during the iteration process.	
Communication of Solution (X1)	Solution is difficult to understand as communicated and is presented in an illogical manner.	Solution is communicated adequately, and thoughts are somewhat organized and/or concise.	Solution is communicated in an organized, clear, and concise manner.	
References and Resources (X1)	There are few references listed, and/or references listed show little relevance to the project's goal.	There are a sufficient number of credible references listed.	Many credible references are listed, reflecting research in the areas covered.	
Resources and AI Reflection (X1)	Form is incomplete or vague and does not clearly explain whether or how AI was used, showing little connection to the submission or understanding of effectiveness and ethics.	Form is complete and clearly explains AI use or the lack thereof with connections to the submission or theme/topic (when applicable), addressing effectiveness and ethical considerations with adequate analysis.	Form is complete, concise, and specific, critically evaluating how AI or non-AI tools shaped decisions and outcomes, with strong alignment to the submission or theme/topic (when applicable) and ethical reasoning.	
			TIER 2 – DOCUMENTATION PORTFOLIO SUBTOTAL (110 points)	

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

PRELIMINARY SUBTOTAL (150 points)	

SEMIFINAL DEMONSTRATION/PRESENTATION/INTERVIEW (50 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Organization (X1)	Team seems unprepared and unorganized for the presentation, with an illogical explanation of the project.	Team is prepared for the presentation and is somewhat organized; team's presentation thesis is, for the most part, logical and/or clear.	Team's presentation with judges is well organized; interview is concise and logical, with a clear explanation of the development of the project.	
Knowledge (X1)	Team members seem to have little understanding of the concepts in their project; vague interview answers are provided.	Team members have a generalized understanding of the concepts discussed and answer questions adequately.	Evidence is clear that team members have a thorough understanding of the concepts discussed; they answer questions thoroughly.	
Delivery (X1)	Communication of the solution is unclear, unorganized, and or illogical; leadership and/or 21st century skills are not evident.	Communication of the solution is somewhat logical and clear; leadership and/or 21st century skills are somewhat evident.	Interview provides a clear, concise, and easy-to-follow analysis of the solution; leadership and/or 21st century skills are clearly evident.	
Solution Demonstration (X1)	Demonstration is unclear or ineffective; solution does not function as intended.	Demonstration is generally clear; solution functions in a basic manner.	Demonstration is clear, organized, and effective; solution functions as intended.	
Team Participation (X1)	Only one person in the group communicates with judges; there is little or no participation from other team members.	Team members all participate to some extent and generally seem to understand the concepts.	Team members seem to fully understand the concepts and share an equal role in the presentation.	
			SEMIFINAL DEMONSTRATION/PRESENTATION/INTERVIEW SUBTOTAL (50 points)	

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

SEMIFINAL SUBTOTAL (50 points)

To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary. **TOTAL** (200 points)

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

ENGINEERING DESIGN EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

A. Event coordinator

B. Judges:

1. Preliminary round, two (2) for each thirty (30) entries
2. Semifinal round, two (2)

MATERIALS

A. Coordinator's packet, containing:

1. Event guidelines, one (1) copy for the coordinator and for each judge
2. TSA Event Coordinator Report
3. Stick on labels for entries

B. One (1) stopwatch per team of judge

C. Measuring tape

D. Display tables for entries (minimum depth 18")

E. Interview questions

RESPONSIBILITIES

AT THE CONFERENCE

A. Attend the mandatory coordinator's meeting at the designated time and location.

B. Report to the CRC room and check the contents of the coordinator's packet.

C. Review the event guidelines and check to see that enough personnel have been scheduled.

D. Inspect the area(s) in which the event is to be held for appropriate set-up, including room size, tables, chairs, etc. Notify the event manager of any potential problems.

E. At least one (1) hour before the event is to begin, meet with judges and assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.

F. Ensure the judges have access to the online judging system.

EVENT CHECK-IN

A. Check in the entries at the time stated in the conference program.

B. If check-in is on the first day of the conference, requirements for attire do NOT apply during check-in.

C. Late entries are considered on a case-by-case basis and only when the delay is caused by events beyond participant control.

D. In order to compete, participants must be on the entry list or must have CRC approval.

E. Place an entry number on each display and prototype/model.

F. Instruct participants to position displays for viewing.

G. Secure the entries in the designated area.

PRELIMINARY ROUND

A. Judges independently assess the entries:

1. Initial round of judging scores the Display and Prototype/Model criteria of the entries to determine the top twenty-four (24) participants.
2. Second round of judging scores the documentation portfolios of the twenty-four (24) identified participants based on the initial round of judging to determine the twelve (12) semifinalists.

B. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and CRC manager to determine either:

1. To deduct twenty percent (20%) of the total possible points in this round or
2. To disqualify the entry

The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.

C. Create semifinalist sign-up sheet for each team's final presentation.

SEMIFINAL ROUND

A. Semifinalist teams report at the time and place stated in the conference program to sign up for a demonstration/presentation/interview.

B. Semifinalist teams report at the assigned time and place for the demonstration/presentation/interview.

C. Manage the completion of the on-site demonstrations/presentations/interviews.

D. Discuss rule violations (e.g. 20% deduction, disqualification) and have all relevant parties initial the rating form.

E. If necessary, manage security and the removal of materials from the event area.